MILITARY SPECIFICATION SHEET

Light Set, Dental Operating, Field

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the light set described herein shall consist of this specification sheet and the latest issue of MIL-D-42048 General Requirements for Deployable Medical Systems.

REQUIREMENTS

Shall be a field dental operating light set for direct and simple attachment to NSN 6520-00-181-7349 Chair and Stool Unit, Dental Operating Field. The light set shall be suitable for field use as a source of illumination during dental procedures. The set shall provide a total operating lighting system. The light set shall consist of a dental reflector-type operating light, a post (2 piece), chair mount bracket, a post hub, mounting hardware, cable assembly (extension power cord), replacement lamps, service manuals, templates, mounting instructions, and tools packaged in a suitable reuseable container.

The set shall include the components listed in table I, in the quantities specified therein, and shall meet the requirements specified herein. The post (lower post and extension post), chair mount bracket, and post hub shall be in accordance with Defense Personnel Support center (DPSC) Drawing 25433. DPSC Drawing 2584-073, Floor Plate shall apply for template hole locations.

TABLE I. Field Dental Operating Light Set.

ITEM	COMPONENT	QUANTITY (EACH)
1	Light, Dental, Pelton and Crane Light Fantastic II, or equal, (Pelton and Crane Aztec tan color, or equal), with 6 foot (minimum) supply line cable as specified herein,	1
	As an alternate,	
	Light, Dental A-dec 6300 Light, or equal, (A-dec tan color, or equal), with 6 foot (minimum) supply line cable as specified herein.	

AMSC N/A FSC 6520

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TABLE I. Field Dental Operating Light Set (continued).

ITEM	COMPONENT	QUANTITY (EACH)
2	Bracket, Chair Mount (3-Hole Mount) (DPSC Drawing 25433)	1
3	Hub, Post (DPSC Drawing 25433)	1
4	Set Screw, 3/8-16 UNC-2A X 3/8 inch, Hexagon Socket, cup point	4
5	Screw, 1/4-20 UNC-2A X 1-1/4 inch, Hexagon Socket Cap	4
6	Lockwasher, Spring (for 1/4 inch Screw, Item 5)	4
7	Post, lower (28 inch) (DPSC Drawing 25433)	1
8	Post, Extension, 25 inch Overall (DPSC Drawing 25433) (21 inch Plus 4 inch Collar)	1
9	Lamp Replacement Package (1 Bulb, Instructions & Tool, if necessary)	2
10	Thumb Screw, 3/8-16 UNC-2A X 3/4 inch (flat point)	4
11	Screw, Flat Head (Countersunk), Hexagon Socket Cap, 3/8-16 UNC-2A X 1-3/4 inch	4
12	Nut, Hexagon, 3/8-16 UNC-2B (For Item 11)	4
13	Flat Washer, 3/8 inch (For Item 11)	4
14	Lockwasher, 3/8 inch, Spring (For Item 11)	4
15	Cable Assembly, Extension Power, 12 feet ± 6 inches, Molded, 14 Gage Stranded, Copper Wire, 2 Pole, 3 Wire Grounded, Thermoplastic Insulation, Molded Connectors, 3 outlet (female) molded receptacle at one end, complying with UL requirements for outdoor use	1
16	Wrench, Combination, Open End and Box, 9/16 inch Across Flats (for 3/8 inch Nut or Bolt)	2
17	Key Wrench, Socket Screw, L-Type, 3/16 inch (for items 4 & 5)	2
18	Key Wrench, Socket Screw, L-Type, 7/32 inch (for item 11)	2
19	Container, Reusable Packing, Hardigg Industries Part No. AL 3018-0905-5949-1, or equal, 4 handles, (2 on each long side, one in each cavity)	1
20	Protective Inserts (reusable, molded cushioning material-for securing, cushioning, and protecting components in the container)	-
21	Manuals, Service (Operator, Maintenance, Repair, & Packing)	2
22	Templates (one each for left and right installation of Bracket) and mounting instructions	2

The light shall be the Pelton and Crane Light Fantastic II, or equal, or A-dec 6300 Light, or equal, for design and performance, having the following salient characteristics:

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The light shall provide a beam of light of shadow-reducing, color-corrected, heat-reduced quality, and shall illuminate the oral cavity in a manner suitable for use in the performance of dental operative procedures. The dental light framework (including arms) shall be tan (Pelton & Crane Aztec tan, or equal), or tan (A-dec tan, or equal).

The light head shall have a lightweight aluminum cast frame finished with baked-on enamel. It shall consist of the glass reflector, the lamp focus bracket assembly, and shall be provided with handles for ease in positioning. The light head shall be mounted in a yoke to permit rotation in a horizontal plane through a range of 240 $\pm$ 10 degrees and vertical rotation of 110 $\pm$ 5 degrees above the horizontal to 170 $\pm$ 5 degrees below the horizontal in the other direction. As an alternate, the light head shall be mounted in a yoke to permit rotation of 200 $\pm$ 10 degrees and a minimum rotation of 30 degrees above the horizontal to a minimum of 120 degrees below the horizontal in the other direction. The yoke shall be a lightweight aluminum casting (or as an alternate, formed steel channel) finished with baked-on enamel and shall contain the on-off switch and the concealed wiring to the lamp. The on-off switch shall be a paddle-blade toggle switch.

The reflector shall be small, fabricated of glass, not to exceed 7 3/4 inches wide and 9 inches high. It shall have a shape designed to make efficient use of the cylindrically-shaped filament. The reflector shall have a continuous varying contour which, in effect, shall be equal to an infinite number of flutes. The reflector shall reflect the image of the entire filament from no less than 180 degrees of the reflector vertical curvature. The curved glass reflector shall incorporate pattern color correction and heat reduction. The reflector shall be coated on the front surface with a dichroic coating which shall reflect cool, white, color corrected light, but shall permit infrared energy to pass through, thereby reducing the temperature at the patient. The back outside coating shall be a ceramic material (frit) which shall be fused into the glass. This coating shall diffuse the light emitting from the back of the light.

The source of the light rays shall be from a bulb positioned in the middle of the reflector, shielded from view in front and the individual light rays shall cross in both directions. The light head shall be supplied with a clear, heat-resistant polycarbonate (General Electric Lexan, or equal) shield having a minimum thickness of .040 inch. The light source shall be a single tubular, bromine cycle, quartz-halogen lamp rated at 150 watts, 25 volts (General Electric Q150T4/CL/PC, or equal). As an alternate, the light source shall be a round, quartz-halogen, xenon-cycle lamp with a single-ended, double-contact, pin type base. The bulb shall be Osram HLX64640, 24 volt/150 watt, or equal. The service life of the bulbs shall be no less than 2000 hours at the high intensity setting. The bulb shall maintain its

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lumen output throughout its life and shall have self-cleaning properties. The lampholder shall be designed for use with quartz-line lamps and shall be suitably rated.

The reflected beam shall be a singular, concentrated beam so as to produce a consistent, even, rectangular pattern measuring at least 3 inches high and 6 inches wide at 27 inches.

The intensity shall be controlled with a three-position dimmer switch on top or bottom surface of the rear arm. At a line voltage of 117 volts, AC, the three-position switch (variable control) shall be set so as to provide the intensity of the light of 1500, 2000, and 2500 footcandles, with a plus tolerance of 300 footcandles, at 27 inches. The light shall meet the intensity requirements specified hereinafter.

The temperature of the beam shall not exceed 8 degrees Fahrenheit in excess of ambient temperature when tested as specified herein after temperature stabilization with the light at its highest intensity.

The color-corrected temperature shall be 3800 to 4700 degrees and shall meet the color requirements specified herein. The distribution curve shall approximate a black body distribution curve of the color temperature so that tissue colors shall appear normal.

The following color and intensity requirements shall apply with a line voltage 117 volts, AC (23, plus or minus 0.5 volts, AC, or as an alternate, 15, plus or minus 0.5 volts, AC across the lamp). The light shall be focused on a target 28 inches from the center of the lamp and shall be in focus, free of striations and shadow lines for all of the target distances specified hereinafter. The intensity switch shall be at highest setting.

All reflected light to be of a color defined by a box on the Commission Internationale De L'eclairage (CIE) chromaticity diagram which has coordinates noted in table II below:

TABLE II. Coordinates defining color of reflected light.

Coordinate	X	* Alternate	Y	* Alternate
1	.368	.398	.365	.410
2	.398	.360	.380	.395
3	.410	.360	.416	.375
4	.375	.390	.400	.382

*Alternate values apply to alternate voltage values (15 ± 0.5 volts).

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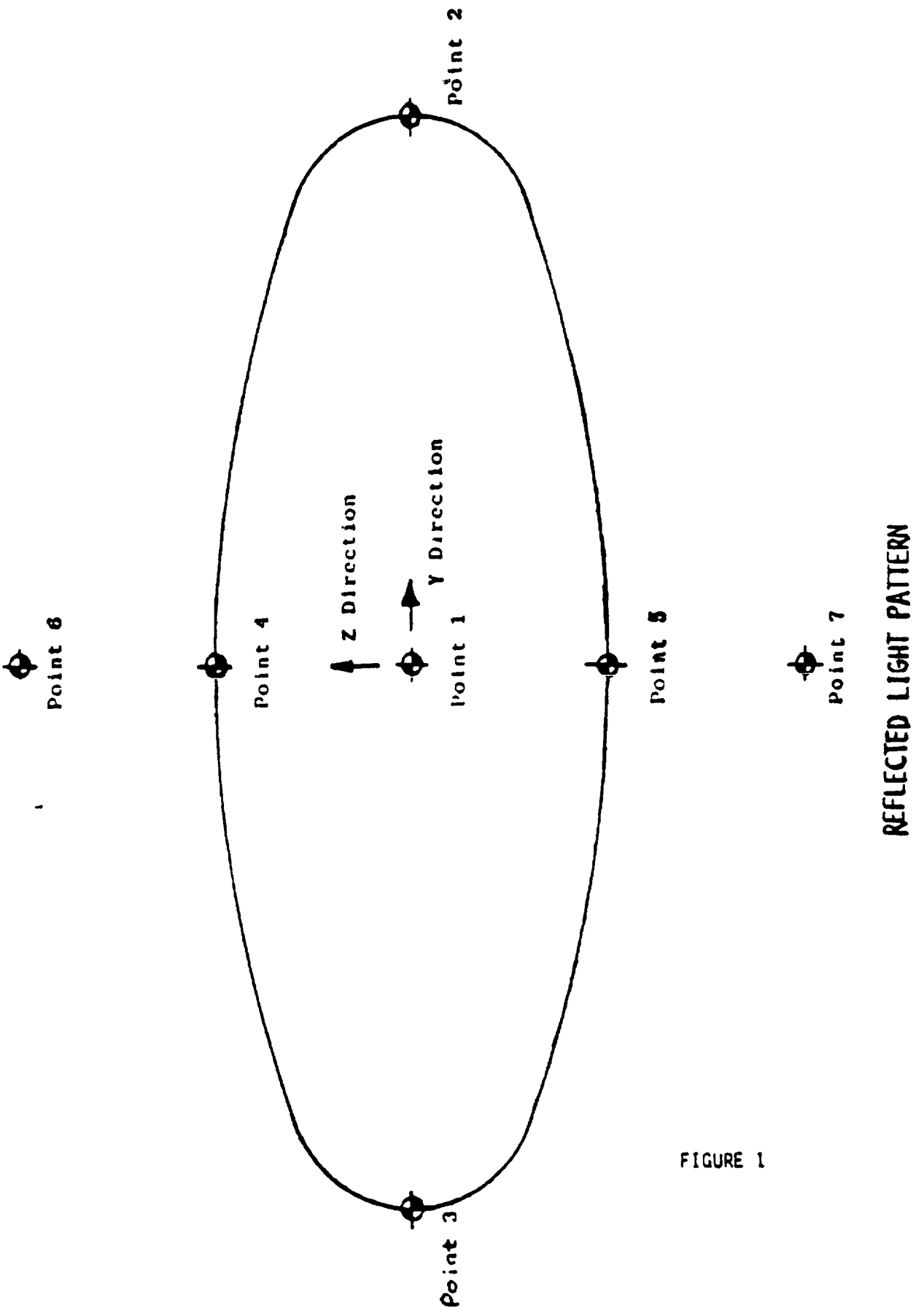


FIGURE 1

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The maximum deviation in color between any two areas of reflected light on the same reflector shall have a maximum variation of .01 as measured on the CIE chromaticity diagram.

The minimum intensity of the light at the center of the pattern (point 1 of Figure 1) shall be as shown in table III below:

TABLE III. Minimum intensity levels.

Distance from Center of Lamp to Target in Inches	Minimum Intensity at Center of Pattern in Footcandles
36	1700
30	2000
24	2500
18	2500

The following maximum and minimum intensities shall apply at all target distances from 18 to 36 inches:

The maximum intensity of the light pattern at a distance of 3 inches from the center of the pattern in the Z direction of Figure 1 (Points 6 & 7) shall be 5% of the intensity measured at the center of the pattern.

The minimum intensity of the light pattern at a distance of 1-1/2 inches from the center of the pattern in the Z direction of Figure 1 (Points 4 & 5) shall be 40% of the intensity measured at the center of the pattern.

The minimum intensity of the light pattern at a distance of 4 inches from the center of the pattern in the Y direction of Figure 1 (Points 2 & 3) shall be 40% of the intensity measured at the center of the pattern.

There shall be no visually apparent hot spots or shadows within the pattern at all target distances from 18 inches to 36 inches. There shall be no complete pattern separation at all target distances from 18 inches to 36 inches.

A long life step-down transformer shall provide proper voltage to operate the quartz-line type lamp. The transformer shall isolate the entire system from normal line voltage to eliminate shock hazards.

The yoke of the lamp head shall be mounted on the front arm which shall be counterbalanced by an internal gas spring or an adjustable compression spring mechanism to compensate for the weight of the lamp head and to prevent vertical drift of the front arm from the selected position. The front arm shall enclose the wire leading from the transformer to the light bulb.

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The movement at the base of the rear arm shall be full 360 degree rotation. The arm shall be 19 ± 1 inches long and shall enclose the wire from the light to the transformer. The distance from the center of the rear post to the centerline of the light head shall be 52-1/2 ± 1/4 inches when the arms are fully extended.

The light shall have a 6-foot, minimum, continuous exposed power cable (free length which feeds through the entire length of the light post, both pieces, and extends beyond the light for a minimum of 18 inches for connection to the cable assembly), fully insulated, heavy duty 3-wire grounded, electrical supply cable furnished with a plug that shall meet specification requirements (suitable for field use).

The upright post shall consist of two pieces, a short (lower) post and an extension (upper) post. The lower end of the short post shall fit into the post hub which is attached to the chair mount bracket. The extension post shall have a reduced diameter portion which shall fit into the short post and sized at the opposite end to receive the adapter of the rear arm of the dental light. The posts shall be chrome-plated, satin buff finish.

The chair mount bracket shall be suitable for mounting on the plate (floor plate) of NSN 6520-00-181-7349, Chair and Stool Unit, Dental Operating, Field. The post hub shall be suitable for attachment to the chair mount bracket and shall accommodate the lower end of the short post and shall provide for attachment by screws (specified in table I).

A 12-foot ± 6 inches extension integrally molded power cable assembly (extension cord) shall be furnished. The cable assembly shall be 14 gage (minimum) stranded copper wire, 2 pole, 3 wire grounded, thermoplastic insulation, with a 3 female outlet, straight line, receptacle at one end. Receptacles and plug shall meet specification requirements. The cable assembly shall comply with UL requirements for outdoor use.

Two replacement lamp packages shall be furnished. The package shall include a replacement bulb, instructions, and required tools (if necessary) to facilitate bulb replacement.

Tools shall include the following:

Two 3/16 inch and two 7/32 inch (total of 4) hexagon socket head screw key wrenches, L type, short series. The wrenches shall be treated or coated to be rendered corrosion-resistant.

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Two open end and box combination wrenches (open end and offset box opening). Wrench opening, 9/16 inch across flats (for 3/8 inch nut or bolt). Wrenches shall be treated or coated to be rendered corrosion-resistant.

The light set (including all components) shall be securely contained in a reusable container which shall be lightweight and have reusable protective inserts. The container shall be Hardigg Industries Part No. AL3018-0905-5949-1, or equal, for design and dimensions. The case shall have 4 handles (2 on each long side, one in each cavity).

The inside dimensions shall be 29-3/4 inches in length, 17-7/8 inches in width, 14 inches in height. The outside dimensions shall be 32-7/8 inches in length, 21 inches in width, 16 inches in height (base 10-5/8 inches, lid 5-3/8 inches), maximum outside volume 6.4 cubic feet. Tolerances on dimensions shall be $\pm$ 1/8 inch. The container shall nominally weigh 32 pounds (empty). The color of the case shall be No. 34087-Olive Drab (Lusterless) in accordance with FED-STD-595A.

The protective inserts shall be of formed configuration to accept all components. The inserts shall secure and cushion the light set and all components in the container. The inserts shall be reusable and protect against damage in transit such as by movement, vibration, shock, temperature environment.

Finish. The finish of all components of the set shall be suitable for field use and shall be suitable for the purpose intended.

Material. Materials used shall be of the highest quality throughout and shall be entirely suitable for the purpose intended.

Unit. Set (SE). One light set, as specified, constitutes one unit. Each light set shall contain template and chair mounting instructions.

QUALITY ASSURANCE PROVISIONS

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification data sheet where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used in this specification data sheet, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

First article inspection. First article inspection shall be performed on the light set when a first article sample is required. This inspection shall include examination and tests.

Quality conformance inspection. Quality conformance inspection shall include the examination and tests specified herein.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the component/materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government, and provided also that the test methods and the results obtained comply with those described in the specification data sheet.

The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification data sheet requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one light set.

TABLE I. Sampling for examination.

	Inspection Level	AQL (percent defective)
Major A	II	1.0
Major B	S-1	2.5
Minor	II	2.5

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Sampling for tests. Sampling for tests shall be conducted as follows:

a. First Article. When first article samples are required, all tests shall be performed on the first article samples. Failure of any test shall be cause for rejection of first articles.

b. Acceptance testing. Two units representative of each lot shall be subjected to the tests specified herein to determine compliance with specification requirements. Failure of any test shall be cause for rejection of entire lot.

Examination. The light set shall be examined to determine compliance with the requirements of this specification in accordance with Table II. Examination shall not be limited to the possible defects listed herein.

TABLE II. Categories and defects.

Categories	Defects
Major A	
101	Design and construction not as specified.
102	Components of the light set missing.
103	Components, mating parts loose, not properly aligned and fitted.
104	Components fractured, broken, damaged, dented or not properly joined.
105	Components not free from defects which may impair serviceability.
106	Cushioning material missing or damaged.
107	Components not placed, secured and cushioned for protection against damage in transit and field use.
108	Finish not as specified.
Major B	
151	Dimensions not as specified.
Minor	
201	Identification marking missing, incomplete, illegible, or not correct.
202	Caution plates not permanently secured.
203	Manuals missing.
204	Components not free of corrosion.
205	Color of case not as specified.

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Tests. Tests shall be conducted to determine compliance with specification data sheet requirements. Tests shall include: pattern size, movement capabilities, color, temperature rise and intensity. Test measurements shall be performed using accurate instruments of standard make and model commercially acceptable for the intended purpose. The rise in temperature shall be made at the center of the pattern at a distance of 36 inches using a mercury bulb thermometer or a suitable calibrated thermocouple.

MAINTENANCE SUPPORT

Provisioning Technical Documentation. The provisioning technical documentation which forms the Appendix of DPSC-DEPMEDS-AT (DM) is required as indicated on the DD Forms 1949-1, 1949-2, 1423 and 1664. These requirements may be altered or deleted by the solicitation or contract applicable to a particular procurement.

Repair Parts Support Kits. The contractor shall furnish 3 different repair parts support kits. These kits shall be designated "A", "B", "C". Contractor must also furnish technical documentation necessary to identify and determine the range and quantity of support items to be purchased with the end item as spares, repair parts, special tools, and test equipment. The technical documentation requirements are described in the DoD-DEPMEDS-AT (DM).

This specification sheet covers the requirements for the following item(s):

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-01-205-2289	LIGHT SET, DENTAL OPERATING, FIELD, 115/230 volt, 50/60H_z, AC

Custodians:
Army-MD
Navy-MS
Air Force - 03

Preparing Activity:
DoD-MB

Agent:
DLA-DM

Project No. 6520-2065

Location: NCR Disk #9

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-42048/1	LIGHT SET, DENTAL OPERATING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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